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Introduction

In the past few centuries, the microscope has advanced from a single hand-ground lens to modern light-sheet fluorescence microscopy. The evolution allowed for further advances in cell tracking research, to which instruments can now image an entire living zebrafish embryo in three dimensions over time at single-cell resolution.[1] This has produced "digital embryos" that record the position, movement, and division of tens of thousands of cells throughout development.[2] Although acquiring these images has become much easier, the bottleneck has now shifted to analysis.

A 3D-time dataset contains thousands of cells that are visually similar. They move, deform and divide which can be difficult to track without precise analysis techniques. Turning the raw signal into accurate cell lineages is still tedious work that demands massive manual effort. This project aims to fill that gap by developing algorithms to detect cells in 3D-time microscopy of zebrafish embryos, link them across time, and identify division events to reconstruct their lineages. As the description of the competition this term project is a part of states, this research will help remove a manual bottleneck in developmental biology and accelerate research that depends on quantifying how cells behave.

Data Preparation and Approach

Before a model can be chosen, the raw imagery data should be understood and reshaped to the structure that the task is scored on. One 3D volume is comprised of 64-z slices at each time-point, and the 100 time-points make up the animated visualization of cell movement. Any distance computed should be scaled into physical microns rather than in it’s initial units of measurement; voxels. Leaving the measurement format as is would badly distort the distance for the cells for tracking. According to

the data description writeup, the ground truths are provided as `.geff` graph nodes and edges. These nodes consist of a cell's centroid at one time-pont, stored as integer **t**, **z**, **y**, and **x** coordinates. Edges are made of links joining the same cell between consecutive frames. For example, a single cell tracked across 20 frames has 20 nodes joined by 19 edges. At division, one nodes has two outgoing edges, branching into two daughter nodes. One important note is that only some cells in each frame are labeled, which the scoring metric accounts for.

Model Choices and Reasoning

As a bassline model, blob detection algorithm Laplacian of Gaussian (LoG) with linear assignment linking will be used. Use of this particular version of blob detection over others such as Difference of Gaussian (DoG) or Determinant of Hessian (LoG) is preferred due to the high accuracy and speed [4] on small blobs seen in the microscopy data. LoG and linear assignment will work in two stages: the blob detection will first return a list of cell centroids. It has no knowledge of the time-points or the frames that connect them. It's job is to simply locate the cells in a frame. Linear assignment is the linking part of the process that takes the centroids from LoG in from *t* and *t+1*, then decides which point in *t* corresponds to which point in *t+1*. What sits between these two stages is the *cost matrix*, which is used to quantify and compare the distance between the *frame+t* detections, as rows, and the *frame-t+1* detections as columns. Small distances indicate that the two microns are likely the same cell. Then, linear_sum_assignment finds the one-to-one paring of rows to columns that minimized the total distance. This is a standard practice for cell tracking that has been used in bioinformatics research,[5] [6] which makes it best fit for a model to compare the other ideal and experimental models with.

Two contestants for an "ideal" model to use, which is suspected as the best fit for this task, are StarDist and Cellpose. StarDist casts rays outward from a predicted center and predicts the distance to the boundary along each fixed-angle ray [7]. The "Star" part of the name refers to the star-shaped polygons, which can draw a straight link to every point in it's boundary without crossing outside the shape. However, if the cell is genuinely irregular then StartDist can't represent it since a star-convex

polygon has no way to describe a boundary that curves back on itself. Cellpose takes a more generalized approach that makes no assumption about the cell shape and instead respects the observed shape of a cell [8], tracing an irregular boundary faithfully. It predicts a flow field that points every pixel toward its cell's center, plus a per-pixel probability of being inside a cell; pixels whose flows converge on the same center form one cell, letting the boundary emerge rather than be imposed.

Doing some research on other competition contestants answers, it was discovered that there are indeed irregularities within the cell data. For the sake of caution, Cellpose will be used since it can better trace irregularities. Something to keep in mind while using Cellpose though is that it is sensitive to its diameter parameter, which is the assumed cell size. Setting this hyperarameter wrong is the most common cause of over-or under segmentation.

For the experimental model, an unsupervised learning model called JEPA, or “Joint Embedding Predictive Architecture” will be used. It learns by predicting masked or future parts of the input in a learned representation state, rather than restructuring raw pixels [9]. This separates it from a different family of self-supervised models called “masked auto-encoders”. For this project however, what separates it most from the other models used is the fact that it is an unsupervised approach. The specific application of JEPA for this purpose is the dense-feature variant, which is best fit for the purpose of cell detection because it is explicitly designed to learn dense, specially precise and temporally consistent features [10]. One consideration on using this model for the competition is whether the sponsor, BioHub is in favor of non-commercial weights being used as part of a winning submission even though the model itself is publicly available.

Evaluation Metrics

For every time-point, a chosen model must produce a pair of nodes, one per detected cell, given as a centroid coordinate. For every pair of consecutive time-points the model should produce edges linking each cell to its continuation in the next frame. A division of a cell is represented as a node with two outgoing edges, which represents one parent branching out into two daughters. Reconstruction of

the branch points is half the evaluation; the other is edge accuracy. Edge accuracy represents how well the predicted temporal links match the ground-truth links. In the metric description's terms, the two halves are the "Edge Jaccard" and the "Division Jaccard".

The word "Jaccard" comes from the statistic used to measure similarity between sample sets [3], while "edge" and "division" distinguish between linage between cells and the branching out of cells through mitosis respectively. A predicted edge counts as correct (a true positive) when both of it's endpoints match ground nodes within the 7.0 μm tolerance, and the two ground truth nodes are connected by a ground-truth edge. A predicted division counts as correct when, for a given ground-truth division, the predicted graph contains a connected component that covers the pre-split (parent) stage and touches both daughter lineages.

The Division Jaccard evaluates the branching out of the cells, checking whether each ground-truth division shows up in a predicted graph as a component that covers the parent and reaches both daughters. A division event is the split of one cell into two through the process of mitosis, by which a parents cell gives rise to two daughter cells. This is the branch point that lineage reconstruction needs to capture. In summary, the detection feeds both the Edge and Division Jaccards. They score the links between cells over time with the edge Jaccard and splits of one cell into two with the division Jaccard. An important note to remember is that these scores are not weighted identically. The edge Jaccard is weight-averaged per sample, by that sample's (TP+FP+FN), while division Jaccard is micro-averaged across all samples.

Takeaways

By the end of this project, there should be a clear understanding on which of the three models is best at tracking, linking and predicting the movement of zebrafish embryo cells. First using blob detection as a bassline, then Cellpose as a supervised model, and the dense-feature variant of JEPA as an unsupervised model. Then evaluating each of the model's output nodes with the Division and Edge

Jaccards will tell which performed the best. After the results of the best performing model are satisfactory, they will be submitted to the Kaggle competition in the .csv format required.

As stated at the beginning of this proposal, much of the work around microscopy imaging research transitioned from collecting the data, to analyzing it. This project will hopefully contribute to the widening of that bottleneck for scientists in the field.

Risks and Ethical Implications

Since this research isn't based on human subjects, the main consideration for ethical practice would be ensuring the humane collection of cell data from zebra fish. Cell data can obtained from these organisms without killing them and there are agencies and organizations that exist to prevent this from happening.

A small risk involved with the submission of a solution is if the JEPA model performs best. In this scenario, the competition rules offer leeway in that pretrained models carrying an incompatible (non-commerical) license do not have to be open-soruced as part of a winning submission. However, a larger technical feasibility risk is getting V-JEPA to work at all the constraints at all. Even if JEPA is able to classify the cells, it produces features, not locations of links. A pipeline with a detection head, linker and division logic bolted on would be required for this implementation. For reference, the other two models used have different detection methods but will share the same linking logic, assigning detection across frames via cost-matrix + Hungarian. Additionally, the division logic and output formatting should also be identical between the first two approaches; LoG + Linear Assignment linking and Cellpose.

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